

SQL FUNDAMENTALS: SELECTING & FILTERING

SQL EXERCISE 1

1. RETRIEVE ALL COLUMNS FROM THE EMPLOYEE TABLE

SELECT *

FROM employees;

Id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	JHB
2	Brian	Lee	HR	45000	2019-03-22	CT
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	PTA
5	Eva	Naidoo	IT	72000	2017-09-20	JHB

2. FIND ALL UNIQUE DEPARTMENTS

SELECT DISTINCT department

FROM employees;

department
IT
HR
Finance
Marketing

3. RETRIEVE FIRST AND LAST NAMES ORDERED BY SALARY DESCENDING

SELECT first_name,

last_name

FROM employee-table

ORDER BY salary DESC;

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. RETRIEVE THE TOP 3 HIGHEST PAID EMPLOYEES

```
SELECT id, first_name, last_name, salary
FROM employee
ORDER BY salary DESC
LIMIT 3 ;
```

id	first_name	last_name	salary
5	Eva	Naidoo	12000
1	Alice	Green	10000
3	Cathy	Zulu	65000

5. FIND EMPLOYEES IN THE IT DEPARTMENT

```
SELECT id, last_name, first_name, department
FROM employees
WHERE department = 'Finance' AND salary > 60000;
```

id	first_name	last_name	department
5	Eva	Naidoo	IT
1	Alice	Green	IT

6. FIND EMPLOYEES IN FINANCE WITH SALARY > 60000

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'Finance' AND salary > 60000;
```

id	first_name	last_name	department	salary
1	Cathy	Zulu	Finance	65000

7. FIND EMPLOYEES IN HR OR MARKETING

```
SELECT id, first_name, last_name, department
FROM employees
```

WHERE department = 'HR' or department = 'Marketing';

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Morreia	Marketing

8. FIND EMPLOYEES NOT IN IT

```
SELECT id, last_name, first_name, department
FROM employees
WHERE department != 'IT';
```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Morreia	Marketing

9. FIND EMPLOYEES IN IT, HR or Finance using IN

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance')
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. Find employees in IT with salary > 65000 and city Johannesburg

SELECT id, first_name, last_name, department, salary, city
FROM employees

WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	10000	Johannesburg
5	Eva	Naidoo	IT	12000	Johannesburg